

Trusted AI Safety Expert: Module 1

Official Study Guide



Version Number: 20251015

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Introduction to TAISE

If AI is going to shape the world, we need people who know how to shape it safely.

The **Trusted AI Safety Expert (TAISE)** certificate, created by the Cloud Security Alliance (CSA) and Northeastern University, is a rigorous, research-backed program for professionals who build, manage, or audit intelligent systems. Through 10 comprehensive modules and a final certificate exam, learners gain practical skills to evaluate and mitigate real-world AI risks, apply AI safety and security controls, navigate compliance frameworks, and lead responsible AI adoption across industries. TAISE is more than a certificate—it's a commitment to advancing safe, secure, and responsible AI.

Receive a Respected Industry Credential

TAISE is co-issued by CSA, **the global leader in cloud security standards**, and Northeastern University, **a top academic institution for AI ethics and experiential learning**. This joint recognition enhances your professional credibility, providing a globally respected credential to highlight on resumes, LinkedIn, and professional profiles.

Acquire In-Demand Skills

Professionals will learn to work with frameworks such as **NIST AI RMF, CSA AICM, ISO standards**, and **MITRE ATLAS** while mastering **threat classification, data privacy, model governance, and AI-specific incident response**. These capabilities directly map to the urgent needs of enterprises adopting AI.

Go Beyond Theory

The **TAISE Prompt Library** transforms concepts into hands-on practice. It offers curated prompts and exercises that show learners how to use large language models to reinforce and apply safety principles from each module. While not an exam prep tool, it helps professionals **translate training into job-ready skills**.

Module 1: Introduction to AI

In this introduction to artificial intelligence (AI), you will gain an understanding of the essential concepts behind modern AI systems. We will also introduce key terms relevant to cloud computing and AI, providing the foundation for future modules. We begin by examining the philosophical and historical ideas surrounding AI, including early innovations and newer generative models. While questions about intelligence, consciousness, and the distinction between narrow AI and artificial general intelligence (AGI) are largely beyond the scope of this course, awareness of these concepts is important in understanding the challenges of modern AI systems.

We will trace the history of AI through pivotal moments, including the 1956 Dartmouth conference, the development of neural networks, and pioneering work on deep learning in the 21st century. You will learn the difference between traditional AI, which analyzes and classifies existing data, and generative AI, which creates new content. We will then discuss what differentiates large language models (LLMs) from other forms of generative AI. You will explore five major approaches to generative AI: auto-regressive models, variational autoencoders (VAEs), generative adversarial networks (GANs), transformers, and diffusion models. You will see how each of these approaches is relevant for today's generative AI models.

Finally, we will explore cloud service models (IaaS, PaaS, SaaS). We will discuss how each model relates to the implementation and use of generative AI and introduce the security considerations, computational requirements, and ethical implications associated with each service level. You will build on this foundational understanding of AI and cloud service models in upcoming modules.



Access the [TAISE Prompt Library](#), a free collection of curated prompts designed to complement the TAISE training. This optional resource provides example prompts and techniques that show how to use large language models (LLMs) to explore, reinforce, and apply key ideas from each module.

Learning Objectives

In this module, you will learn to:

- Define key terms and foundational concepts in generative AI
- Describe generative AI models (LLMs, GANs, diffusion models, and VAEs)
- Trace the historical evolution and milestones in generative AI
- Differentiate between cloud service models (IaaS, SaaS, and PaaS) and their implications for generative AI

1.2 Module Vocabulary

Agentic AI describes AI systems designed to perform tasks with minimal human intervention by learning the rules needed to solve tasks rather than having them hardcoded.

Artificial intelligence refers to the capability of computer-based systems performing tasks that would be considered evidence of intelligence if performed by humans or other naturally intelligent creatures.

Attention mechanism refers to a technique in transformers that allows models to weigh the relative importance of different data elements within a specified context window, similar to selective attention in human perception.

Autoregressive models generate data sequentially by predicting each new element based on previously generated elements, similar to traditional time series models.

Computer vision allows machines to interpret images and videos.

Deep learning is a form of machine learning that combines multiple neural network layers to analyze data and make predictions.

A **diffusion model** is a generative AI approach that learns to reverse a gradual noise-adding process, starting from pure noise, to generate high-quality images or video.

Dimensionality reduction is a technique used in AI and machine learning to simplify or effectively summarize complex high-dimensional data into more manageable representations while preserving important information.

Discriminative AI is a kind of machine learning where models learn to differentiate between data. When the training data comes with pre-assigned labels, we refer to this as supervised learning; when it does not, we refer to it as unsupervised learning.

An **encoder-decoder architecture** is a neural network design where an encoder transforms high-dimensional input into compact representations, and a decoder expands these back into the original format.

A **feature** is a measurable property or characteristic of data that serves as input to machine learning and AI models. Features are the variables or attributes that models use to learn patterns and make predictions or generate outputs.

Feature engineering is the process of using domain expertise and statistical properties to create summary measures or representations that help AI models learn more effectively.

A **generative adversarial network (GAN)** is a generative AI architecture consisting of two competing neural networks (generator and discriminator) that work against each other to create realistic synthetic data.

Generative AI refers to a kind of machine learning where models learn to create combinations of data and labels.

A **hallucination** refers to factually incorrect, nonsensical, or unrelated outputs generated by AI models, particularly LLMs, due to their reliance on statistical patterns rather than conceptual understanding.

Large language models (LLMs) are AI models trained on vast text corpora using transformer architecture, which are capable of generating human-like text and interacting through natural language.

Machine learning refers to algorithms that enable systems to learn from data without explicit instructions from humans. Machine learning is a kind of AI.

Natural language processing (NLP) enables machines to process human language.

A **neural network** is a digital system that is created by a series of connected nodes. They are called “neural networks” because the structure is modeled after the human brain.

Principal component analysis (PCA) is a linear dimensionality reduction technique used in machine learning to transform high-dimensional data into a lower-dimensional representation while preserving as much variance as possible. PCA identifies the principal components—directions in the data where variance is maximized—and projects the original data onto these new axes.

Regression models are a type of supervised machine learning algorithm used to predict continuous numerical values based on input features, i.e., data. Regression models in AI learn the relationship between input variables (features) and output variables (targets) from training data.

Reinforcement learning with human feedback (RLHF) is a training method where human evaluators provide feedback on AI outputs to guide models toward generating more accurate and appropriate responses.

Retrieval augmented generation (RAG) is a technique that links LLMs to external databases or document corpora to reduce hallucinations by grounding responses in factual information.

Stochastic parrot is a metaphor describing how LLMs generate responses based on statistical patterns without genuine comprehension, similar to parrots mimicking speech.

A **token** is the fundamental building block of text in LLMs. It represents broken-down pieces of text that the models process and generate.

The **transformer architecture** is the neural network architecture introduced in 2017 that uses attention mechanisms to process sequential data in parallel, enabling the creation of modern large language models.

Variational autoencoders (VAEs) are a type of generative model that uses cooperative encoder-decoder neural networks that learn latent representations of data to generate new synthetic outputs.

Overfitting occurs when an AI or machine learning model learns the training data too well, including its noise and peculiarities, resulting in poor generalization to new, unseen data. An overfitted model essentially "memorizes" the training examples rather than learning the underlying patterns.

1.3 Definitions and Concepts of Artificial Intelligence

Key Concepts and Definitions

The field of artificial intelligence (AI) is full of jargon. While most aspects of AI are grounded in technical concepts from computer science, physics, engineering, statistics, and applied mathematics, it is possible to understand the concepts behind AI without need for a deep technical background. Our goal in this module is to provide a high-level description of the key terms and ideas you will encounter on your AI journey. For those of you with a background in mathematical or computational fields, or if you are just curious to learn more, we encourage you to follow up on the concepts described in this section with additional courses or readings.

Discriminative vs. Generative AI

Until recently, the most common kinds of AI were discriminative or descriptive machine learning (ML) models. As discussed in detail below, machine learning—where models learn functions and parameters from data—is a form of AI. In some contexts, these discriminative ML models are also referred to as predictive AI; however, we use the term “discriminative” because generative AI also makes predictions. The goal of these more traditional ML approaches is to construct a model capable of discriminating between data.

For example, imagine you are given a set of emails and told that some of them contain malware sent as an attachment, while others do not. Using this dataset of emails with and without malware, you could train a discriminative ML model to predict whether new incoming emails contain malware as an attachment. Even if you are more familiar with machine learning (ML), you still might not have heard the terms discriminative, descriptive, or predictive because, until recently, the majority of ML models you were likely to encounter were discriminative.

Supervised and Unsupervised Learning

Typically, we divide ML into either supervised, where data have pre-assigned labels, or unsupervised, where data are unlabeled. In supervised ML, where we have labeled data, e.g., emails that we are told either do or do not contain malware, our goal with discriminative models is to take some data x as input that has been labeled with data y and construct a model that tries to predict the labels.

A simple example is using logistic regression to distinguish between emails containing malware and emails not containing malware. For unsupervised learning, we might be given a collection of emails without knowing whether any of them contain malware. Here, we will use clustering approaches to try to determine whether there are distinguishable types of data present, e.g., whether emails with malware all contain similar kinds of language or characters more associated with code. Often, both supervised and unsupervised learning are paired with a technique called “dimensionality reduction,” where methods are used to simplify or summarize the underlying data.

Feature Engineering and Dimensionality Reduction

Effective data summarization approaches are often referred to as “feature engineering” because they utilize expert knowledge on the subject matter or statistical properties of the data to define summary measures. Common approaches to feature engineering range from simple methods, such as scaling the data based on its mean or standard deviation, to the kinds of encoders used in deep learning models, which we will learn about in the upcoming sections. For dimensionality reduction, the key distinction is whether the methods are linear, such as in principal component analysis (PCA), or nonlinear, such as with autoencoders used in neural networks. The details of the approaches above are beyond the scope of this course.

Understanding Generative AI

Now, let’s discuss what is referred to as generative AI (genAI). Generative AI models enable us to make predictions about labels associated with our data, such as whether an email contains malware. They also help us generate new combinations of data and labels that match the properties in our training data, for example, an email that contains a piece of malware. Keep in mind that all of the generative models we will discuss in this course are also types of ML, meaning they are models that learn from either labeled (supervised) or unlabeled (unsupervised) data. Most modern generative AI advances are improvements in the ability to process massive amounts of data in parallel, i.e., performing multiple tasks simultaneously. Our ability to perform such parallel processing on large datasets is driven by advances in model architecture, which is related to concepts such as dimensionality reduction and feature engineering and improvements in hardware, such as GPUs.

Evaluating Model Performance

Whether we’re dealing with traditional discriminative AI or modern generative AI systems, we evaluate model performance along the same four axes:

1. In-Sample Performance

First, how well does the model perform on the data used to train it? We call this kind of performance “in-sample” and measure it using values such as the root-mean-squared-error.

2. Generalization

Second, how well does the model generalize to data not used during training? When training ML models, it is considered best practice to hold out some of your data during training and then measure model performance against that held-out data. Companies will often use what is called A/B testing, where models are deployed to compete in real-world settings, i.e., model A vs. model B, as a way to measure generalizability.

3. Explainability and Transparency

Third, how explainable or transparent are the underlying modeling approaches and training data? If your head is already spinning with all the AI jargon, you can understand why explainability is desirable. Explainability is also related to the fourth axis.

4. Trustworthiness and Responsibility

The fourth axis is trustworthiness, also referred to as responsibility. Here, we are interested in whether the AI reflects the values, ethics, and norms of our organization and society. Evaluating a model for its trustworthiness is possible when the underlying approaches are transparent.

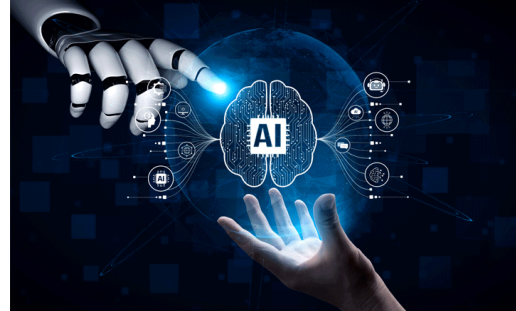
AGI and Beyond

Finally, as discussed in more detail below, AI with human levels of cognition, typically referred to as artificial general intelligence (AGI), is an ill-defined concept and beyond the scope of this course. However, understanding what separates ML from hypothetical AGI and what separates generative from discriminative (often called predictive) AI are important concepts you should understand with confidence after finishing this module.

1.4 How is Machine Learning Related to AI?

How ML is Related to AI

Most of what we call artificial intelligence, including generative AI, is more accurately classified as machine learning. But what is machine learning, and what do we mean when we say a machine learns? In this lesson, we'll explore how machine learning fits within the broader field of artificial intelligence, discuss how machine learning also powers generative AI, and explain how it differs from solving a problem by directly programming the solution into a computer. We'll examine what it means for a model to learn from data, the distinction between supervised and unsupervised learning, and how we assess a model's effectiveness and reliability.



Understanding Machine Learning

Machine learning is a subset of artificial intelligence. Artificial intelligence refers to programs with the ability to simulate human intelligence. What exactly we mean by human intelligence is beyond the scope of this course.

However, one aspect of intelligence is the ability to learn from data, and that's what machine learning algorithms are able to do. They're able to take in data and learn functions or parameters from those data. More precisely, imagine that you have some data—these could be a bunch of emails—and we're interested in building a classifier for whether these emails contain a ransomware attack.

We have some model that we're going to fit to the data; this is the underlying math that we're going to use to actually predict the cyber threats. Then we have something called a loss function, which means how exactly we're going to measure the performance of this model—specifically how we're going to evaluate its performance. You run all these through a training algorithm, and we end up with some trained model that we would call a machine learning model.

A Concrete Example

Here's a more concrete example: Imagine you have some data on threat statistics, you have some data on activity over time, and you have some information on where threats have been detected. In order to build a machine learning model, we have to decide on what kind of mathematical form. We could use a neural network—that's a kind of machine learning.

There's a whole variety of different neural networks. We could use something simpler like logistic regression. Imagine we just have a whole bunch of emails and we've labeled these emails as either 0 (not a threat) or 1 (threat), and we would use a logistic regression model to evaluate and train. Then we would use the appropriate evaluation tool associated with the logistic regression.

In reality, the data are probably not going to be quite as simple, and we might use something like principal component analysis to do dimensionality reduction.

If this concept is unfamiliar, simply imagine that each one of these points is much more complicated than just an email. It might have a whole bunch of other data associated with it, and we would try to identify where there's a split between the threats and the non-threats.

This is something that we could solve with logistic regression. Importantly, this process of principal component analysis is something that machine learning and AI researchers or practitioners would call feature engineering or feature extraction.

It sounds really fancy, but it's basically just taking really high-dimensional data and distilling it down into something that's simpler. That can be important even for a machine learning algorithm.

A machine learning algorithm may not be able to sift through all of the complexity in the real-world data. We might pair it with something, like a principal component analysis, to simplify and filter the data.

What Isn't Machine Learning

One important thing to think about in the context of machine learning is what isn't machine learning. A thing that is not machine learning is standard least squares linear regression. The reason it's not machine learning is that if we're going to do a standard least squares linear regression, we actually know the solution for that—given a bunch of data, like height and weight, we know how to solve for the best fit linear regression to these data.

If a human tells the computer the answer, then it's not machine learning. Machine learning means that the computer has to learn aspects of that answer from the data. This becomes very important when we start to talk about things like agentic AI, because one big difference between agentic AI and more established solutions that may also automate tasks is that the human is not telling the AI agent the solution.

The human is not saying, "Here's how you would classify a cyber threat. You're going to look at these features. You're going to look at the IP address that it came from. You're going to look at some elements associated with the hyperlink that are present and classify it based on the human expert guidance." Instead, if it's an agentic solution or a machine learning algorithm, the algorithm or agent is actually going to learn how to do that process itself.

Types of Machine Learning

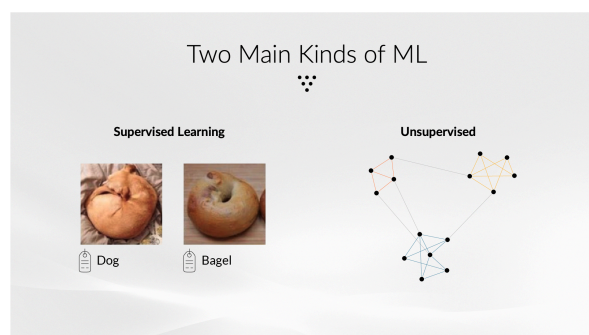
There are two main kinds of machine learning, and it comes down to whether the data are labeled:

Supervised learning: The data are labeled. For example, with dog or bagel images, someone has looked at the images and written down whether it's a dog or a bagel.

Unsupervised learning: We don't actually have labels. We're going to do some kind of clustering to try to identify the groups. Maybe we would find out later that these are the dogs, these are the bagels, and this is something else over here that's neither a dog nor a bagel.

There are lots of different kinds of machine learning models. They are split between whether we have labeled data for supervised learning or unlabeled data for unsupervised learning. Within each of these, we have different approaches:

- Supervised Learning (labeled data)
 - Regression models
 - Decision trees
 - Many other approaches
- Unsupervised Learning (unlabeled data)
 - K-means clustering
 - Principal component analysis



Within each of these, there are subclasses. We have different kinds of regression models and different kinds of K-means clustering. Instead of trying to think about this whole zoo of models, it's useful to think about how it's organized. Whether the data are labeled or not is the most important split.

Then within each of these, whether we're using regression approaches or these tree-like approaches, and in unsupervised learning, whether we're doing some kind of dimension reduction or feature engineering like principal component analysis, or whether we're doing some kind of clustering like K-means clustering—although there's a lot of overlap between how those methods function.

Evaluating Machine Learning Models

How do we evaluate machine learning models? This is going to be true whether it's a generative AI tool that's a machine learning model (generative AI is a type of machine learning), whether it's a neural network, or whether it's a regression model. We evaluate them based on three criteria:

Performance: How well do they perform at the task given the data that they were given—what we might call in-sample performance.

Generalizability: How well do they perform on data that was not included in the training process. We're going to have a whole corpus of emails that we used to train a cybersecurity threat detection model. Then we're going to test that against a bunch of emails that weren't included, or we're going to start running it out in the real world and evaluating its performance as it's seeing new emails come in.

Explainability: How easy is it for someone to sit down and explain to an expert how the tool works. There are some models that even experts can't explain to each other how they work. This has become increasingly important in this AI age. How easy is it to explain to an educated practitioner who's not an expert? How easy is it to explain to a layperson who's never really even experienced one of these tools before or even really thought about the data, like the cybersecurity threats?

Cross Validation

One way that we could formalize this process is through what we would refer to as cross validation. Basically, we want to think about how we test the model in terms of its performance on the data that's used to train the model (the in-sample data) versus its generalizability.

There are many ways we could do that. We could hold some of the data out until the end. We could repeatedly split the data into different sizes and train and test the data. There's a whole variety of ways in which we might go about this.

Importantly, you need to think very carefully about how your data are structured and how the model is going to be used to select the right kind of approach. Ultimately, you have to run these out in the real world to see how they really perform once they're no longer in the training phase but they're actually out there being used.

Using Machine Learning to Detect Cybersecurity Threats

How can we use machine learning to detect cybersecurity threats? As we talked about before, we have our data. We've decided we're going to do a logistic regression—this could also be a neural network, or it could be that we're going to train a custom large language model to identify cybersecurity threats. As I mentioned, we have to pick what's called a loss function. For logistic regression, we know how to do this. Then there's a final step that we might add, which is some kind of penalty associated with model complexity. That's meant to handle this balance between generalizability and in-sample performance. You might have heard of things like regularized regression—that's a form of penalty that we might add to the model. What this ends up looking like is we want to avoid underfitting.

This is the Goldilocks problem. We don't want to make the models so simple that they don't really capture the data. We want to avoid overfitting such that it's so specialized to the data that it was trained on that it's not going to generalize well. Instead, we want to find something right in the middle. To be honest, this is more of an art than a science and requires a lot of very careful thought and a lot of testing out in the real world to get this right.



Key Takeaways

- Machine learning is a kind of AI. It involves training models with data without telling it the answer. That's really important and very important in the context of agentic AI.
- The easiest way to think about organizing all these various models is whether we have labeled data (supervised learning) or unlabeled data (unsupervised learning).
- We evaluate machine learning models based on their performance on the data used to train them (in-sample performance), their generalizability (out-of-sample performance), and their explainability. This explainability piece has become really key in the age of AI.
- It's typically better to have a model that's maybe just a little bit worse but much easier to explain than it is to have a model that's, say, maybe a half percent better but unexplainable. Although you might imagine that there is some context—like maybe you're working in a financial market or you're in some arms race with nefarious actors in a cybersecurity context—where all that matters is performance. It does not matter if you can explain what's going on.
- We also typically want to balance performance and generalizability via this process called regularization, which, in some simple way, just means penalizing more complex models.

Machine learning isn't magic. It's data-driven pattern recognition guided by measures like performance and generalizability. It's also the most common kind of AI, both the more traditional methods and newer

generative AI tools. Understanding more about how machine learning works is key to making responsible, informed use of AI tools.

Generative AI is Still Just Pattern Recognition

A practical and less technical way to think about generative AI is that it is still performing the same kind of pattern recognition as more familiar types of ML, such as logistic regression. However, generative AI models can create (i.e., generate) entirely new kinds of data. At their core, most generative AI models function like autocomplete systems, although, as we will learn, some of the newer image and video generation models work differently. As a result, generative AI models do not understand anything about the underlying data but instead function as high-powered pattern recognition engines.

The autocomplete nature of generative AI is where the term "stochastic parrot" originates, as it resembles the behavior of parrots; generative AI models do not understand what they are saying but respond based on similar past situations. More formally, we might say that most generative AI models build upon much older autoregressive models. These are systems that generate data sequentially by predicting each new element based on previously generated elements. These are models you will have encountered if you've ever studied time series prediction or analysis.

The sequential approach of such autoregressive models underlies many modern AI systems, from language models that predict the next word to music generators that create notes one by one. However, because many current generative AI systems are trained by "masking" labels and developing models capable of predicting the hidden or masked labels, the sequential context is gone, and thus we would not consider them autoregressive. We will return to these ideas in more detail as we define the common kinds of generative AI approaches.

Encoders and Decoders

One of the most important technical genAI concepts is encoders and decoders. An encoder network is a neural network that transforms high-dimensional input data into a compact representation, such as dimensionality reduction paired with feature engineering. A decoder network expands this compressed representation back into the original data format. One of the reasons we believe neural networks perform so well on ML tasks is because they perform built-in feature engineering and dimensionality reduction.

The original encoder-decoder architecture processes input through an encoder, which creates an intermediate representation, and it then generates output via a decoder. Encoder-only models, such as BERT (bidirectional encoder representations from transformers), focus solely on representing input text, and they have historically excelled at classification and question answering. Decoder-only models, such as GPT (generative pre-trained transformer), generate text by predicting tokens—broken down pieces of text representing the fundamental "building blocks" for generative AI—one after the next based on surrounding tokens, making them ideal for open-ended generation tasks. The decoder-only architecture of GPT and other similar models facilitated parallelizable training, which most believe was essential for the scale these models have achieved.

Types of Generative Models

Currently, there are five widely used kinds of generative models:

- I. **Autoregressive models** generate data sequentially, predicting the next element in a series, e.g., a time series, based on a regression model fit to prior data in the series. If you've studied time series analysis, you've likely encountered autoregressive models such as ARIMA. Autoregressive models are the oldest form of generative AI. They are most commonly referred to as machine learning or statistical models, and they pre-date the modern rise of generative AI by decades. At the time, autoregressive models were not called generative AI, but today we would consider them an early ancestor of today's generative AI.
- II. **Generative adversarial networks (GANs)** are composed of two neural networks, a generator and a discriminator, that interact competitively to create labeled data. Imagine the generator network producing some data with labels and the discriminator providing feedback on the likelihood of those combinations, given the training data. The discriminator's output "feeds back" onto the generator, which learns to produce data more likely to pass the discriminator's test.
 - A. A potential application of GANs is the generation of realistic synthetic patient records for cyber security testing without exposing real patient data ([Generative adversarial networks and synthetic patient data: current challenges and future perspectives - PMC](#)).
- III. **Variational autoencoders (VAEs)** are also composed of two neural networks. One neural network learns a latent representation of the data, such as a kind of dimensionality reduction, and another neural network uses the latent representation to generate new data that—unlike GANs—work in concert with each other. VAEs were designed to work on unlabeled data and are primarily involved in unsupervised learning. VAEs introduce a probabilistic approach by encoding inputs into probability distributions, which enables them to generate diverse yet realistic synthetic outputs.
 - A. Potential applications of VAEs include (i.) Anonymizing patient data by generating synthetic representations that retain structural properties without identifiable personal information, which is crucial for secure data sharing and privacy compliance ([Synthetic electronic health records generated with variational graph autoencoders | npj Digital Medicine](#)); (ii.) An intrusion detection model combining VAEs and GANs ([An intrusion detection method combining variational auto-encoder and generative adversarial networks - ScienceDirect](#)); and (iii.) Performing dimensionality reduction on large, complex datasets.
- IV. **Transformers** were developed to model sequential data, such as predicting the next word in a sentence, and evolved from VAEs through supervised learning. They replaced recurrent and convolutional neural networks as the dominant architecture for text data. VAEs are still commonly used for training unsupervised models associated with unlabeled text and other sequential data. Transformers are also composed of one or more neural networks. As discussed above, transformers can be encoder-only (e.g., BERT), decoder-only (e.g., GPT models), or encoder-decoder (as described in the original 2017 paper introducing transformers, "[Attention is all you need](#)").

The key innovation with transformers is a technique called the attention mechanism, where the model learns how to weigh the relative importance of different data elements within a specified context window. The classic analogy is having a conversation with a friend in a crowded room. Your brain is paying selective attention to all incoming signals and filtering out what is effectively “noise” outside of the conversation you are having. Today, most transformers are trained to recognize “masked” or hidden labels using the context around the masked label, where context refers to the combination of data and labels.

As a result, when trained via masking, the ordering of the data is not typically preserved, and thus the resulting transformers would not be autoregressive. Nearly every day, updates are published to transformer-based approaches. Staying current here requires reading blogs, listening to podcasts, and regularly retaking courses reflecting the latest updates.

- V. **Diffusion models** such as Imagen, Midjourney, and Stable Diffusion were inspired by thermodynamic models from physics and are trained in a two-step process where data is corrupted with noise and a model is trained to reverse this noise process. More technically, diffusion models consist of a forward diffusion process that gradually adds the right amount of noise at each step and a reverse sampling process that is trained to remove the noise. The resulting models can generate new data starting from what is essentially white noise.

The underlying architecture of diffusion models is still neural networks. Diffusion models are now considered state of the art for most image and video generation tasks. A number of image generation tools now combine diffusion models with transformers. An example of these hybrid diffusion-transformer models is DALL-E, which we didn’t list above because it was initially a transformer-only model for image generation. Not surprisingly, the enhanced generative performance of diffusion models for images and video has raised [new challenges associated with deepfakes](#), a concept we will revisit in future modules.

1.5 What is “AI”?

Artificial vs. Natural

Calling anything artificial is a way of contrasting it to something considered natural. With that thought in mind, we could answer the question “What is ‘AI’?” by saying that it is something not already found in nature but that nonetheless exhibits aspects associated with natural intelligence. We can immediately see some shortcomings in that definition of AI, namely, that it requires defining intelligence and drawing a line between natural and unnatural. While not useful in practice, it is important to remember that the original goal of AI was to construct machines with human levels of intelligence (at the time considered the only form of natural intelligence). Today, many researchers still aspire to achieve AI with human-level cognition, often referred to as artificial general intelligence (AGI).

Historical and Present Context

Though “AI” as a term is less than a century old (coined during the famous 1956 Dartmouth conference), there are many precursors to the pursuit of human-created machines that demonstrate aspects of intelligence or intelligent behavior. For hundreds of years, the available tools for inventors and tinkerers who wanted to build such “intelligent machines” relied on power coming from water, steam, gears, or electricity, and those technologies also became the metaphors used to describe “natural”—i.e., human—intelligence. AI discussions about intelligent machines now refer mostly or exclusively to digital computer-based machines: Computational metaphors—minds as computers, thinking as computational processes—dominate many present-day discussions of both human and artificial intelligence. We have also come to realize that many other natural systems display intelligence, which has led to a large increase in the study of animal and even plant intelligence.

What is Intelligence?

What is intelligence? And how is it related to other properties, such as being alive, being aware, being self-aware, being sentient, being mobile, communicating through language, having emotions, having a soul or spirit, or being in relation with other intelligences? Not only are these questions beyond the scope of this course but they are highly debated areas of scientific research, lacking the kinds of agreed-upon mathematical and computational answers needed to evaluate AI systems.

Most commonly, AI is understood purely in behavioral terms. Here, an AI system can act or behave in ways that, if a human or other intelligent creature performed similarly, would be evidence of intelligence. The Turing test, invented by Alan Turing, who many consider the father of modern computer science, can be seen as offering a “purely behavioral” conception of intelligence, although that interpretation is also debated. But AI can also be understood in more metaphysical terms. From that viewpoint, an AI system can do things that seem to require intelligence because it is intelligent. Maybe now you’re starting to see the issue associated with defining intelligence.

Moving beyond definitions of AI associated with human behavior, capacities associated with intelligence include perceiving and representing one’s environment, responding flexibly to changes in one’s environment, learning from experience, generalizing to new tasks, and perhaps, most generally, taking action in pursuit of one’s goals. As a research program, more so than as a topic of biological, philosophical, or psychological inquiry, AI pursues ways to create machines that possess those capacities. Indeed, many

AI systems can accomplish many of the previously mentioned tasks, such as sensing one's environment. However, despite the similarities, vast differences exist in how these aspects of intelligence manifest in our artificial machines compared to our natural brains.

Narrow AI vs. AGI

Many paradigms and approaches in AI extend back to the earliest days of computer science. Approaches that focus on particular subsets of intelligence—perhaps for precisely and narrowly defined tasks—are called “narrow AI.” Those aiming at a more general-purpose system are pursuing “AGI” (artificial general intelligence). Often, when someone says AGI, they mean AI with intelligence indistinguishable from what we would consider human but capable of performing super-human levels of computation. Currently, AGI does not exist, and debate continues on whether current technologies are capable of creating AGI. Within each of those approaches, some researchers prioritize building systems that can use and manipulate symbols to represent the world and to solve problems. Others prioritize building systems modeled on the neuronal systems of the brain.

One of the currently dominant forms of AI systems is known as “generative models,” which encompass tools such as Midjourney and chatbots such as Claude and ChatGPT. Generative models are based on deep learning networks, which are based on artificial neural networks. But there is much more to AI than ChatGPT, just as there are many more flying machines than helicopters.

Because there's no precise definition of the term, AI has come to be applied generally to computer or software systems that carry out complex tasks. As a consequence, many systems you might interact with every day are built on technologies that, perhaps just a few years earlier, might have been called “AI” but today go by a different name. What's most important is to determine whether the system has provided specific instructions for solving problems by human experts, something not typically referred to as AI, or whether it has learned the solution from data (a subset of AI called machine learning) or by some other method.

Extend Your Learning



The information below contains readings related to this lesson. These resources are optional and are provided as an opportunity to further your understanding.

- [A very gentle introduction to large language models without the hype](#). Riedl, M. (2023, May 25). Medium. (35 min)
 - This article was written to give individuals with no background in computer science insight into how generative AI, such as ChatGPT, works



- [AI vs Machine Learning](#). IBM Technology. (2023, August). YouTube. (6 min)
 - This video helps clarify the differences and relationship between concepts like AI, machine learning, and deep learning.
- [Anatomy of an AI system](#). Crawford, K., & Joler, V. (n.d.). (27 min)
 - In this article, you'll read about the creation of the Amazon Echo from a human labor and data perspective.

1.6 History of AI Development

Examining AI Origins

To fully appreciate the current state of AI technology, we need to embed that technology in history. We will explore that history in this lesson by considering the theoretical precursors and origins that made discussions of artificial intelligence possible in the first place.

Theoretical Origins of AI

Alan Turing was a mathematician and philosopher who turned to computer science as a logician and codebreaker in World War II. He was a pioneering researcher in computation, which at the time primarily focused on a major idea: that certain decision-making processes, such as ones that yield a “yes” or “no” answer, could be computationally represented through algorithms that resolve to a decision in an automated way.

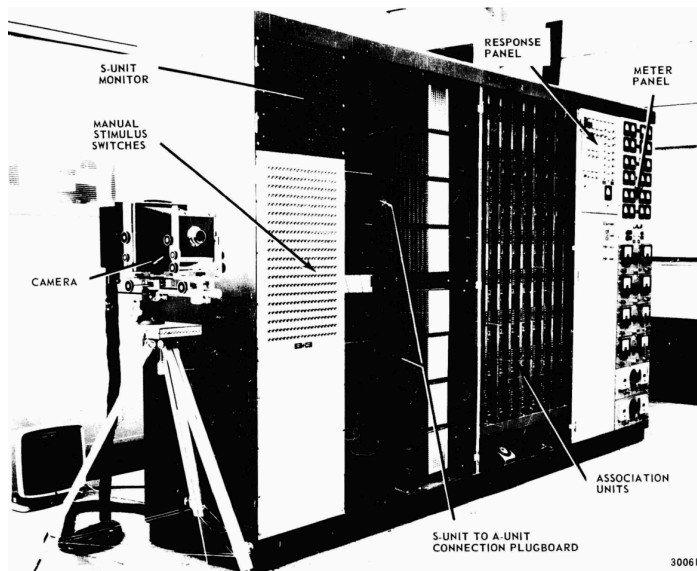
This work was situated in contemporary work in formal logic, including research by Kurt Gödel and Alonzo Church, which aimed to formalize the elements and procedures of logic through abstract mathematical symbols and expressions. Formal logic was an active area of theoretical research by itself, but its significance was amplified when it coincided with another emerging philosophical area of research at the time. That area argued that the human brain as a whole – our thoughts, how we reason, how we make decisions, and how we respond to the stimuli of daily life – could be represented as a series of computations that could be expressed through formal logic.



Dartmouth College

The term “artificial intelligence,” as we know it today, was material to this philosophical discourse. It was first coined by John McCarthy in 1956 at a convening of researchers at Dartmouth College. The Dartmouth Summer Research Project on Artificial Intelligence brought together computer scientists to explore how the complexity of human intelligence could be decomposed into smaller elementary pieces and then replicated in machines, opening many directions of inquiry that laid the basis of AI technological development.





Perceptron

In 1958, this confluence of theoretical origins collided with the invention of the first technology that would be recognized as artificial intelligence: the perceptron. In research conducted with the U.S. Office of Naval Research, Frank Rosenblatt, a research psychologist and engineer at the Cornell Aeronautical Laboratory, built a machine that was trained to “look” (with a camera) at a series of cards with a colored square on the left or right side of the card and differentiate the cards based on the location of the square. The resulting “perceptron” was named as such because it appeared to perceive distinctions between the cards and classify the cards based on those distinctions.

The image above is a figure from the perceptron operating manual

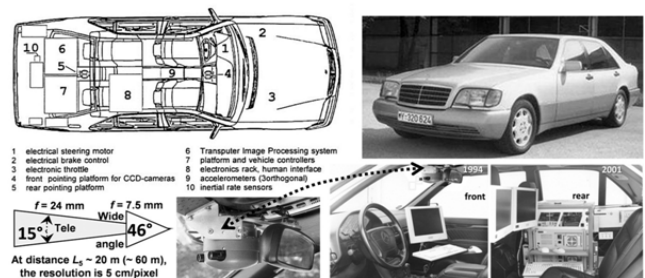
At the time of its unveiling, the perceptron sparked dramatic interest among both researchers and the public. It also would later be considered the first example of a neural network, a specific form of machine learning that is used widely in AI technology today. However, as time went on, public hype over the first evidence of a machine seemingly able to learn was replaced by growing skepticism – Was it even possible that a machine could “learn” like a human does? That’s ridiculous! – and soon thereafter, thrown into obscurity.

AI Winter

Between the late 1970s and early 1990s, research in AI entered a period of dramatically diminished activity, caused by significant pullback in governmental funding for AI research. This “AI Winter,” as it would be called first in 1984, was spurred by a series of reports published in the 1970s that cast doubt on the promise of AI, suggesting that the early advances of AI research had overpromised and under-delivered on the claim to engineer machines to learn in the way that humans do.

First Driverless Car

Throughout the AI Winter, research in AI occurred at a sluggish cadence. The period was driven by a hype and bust cycle that viewed machine intelligence as theoretically interesting but practically unachievable, because the technology required to support ongoing research was not advancing in pace. The AI Winter persisted until new advances in applied AI research reignited public interest in the value of AI. Those advances reopened the AI discussion



because they demonstrated that AI not only was a theoretical exercise but could also have practical value in the real world. For example, one of these in the late 1980s was the development of the first driverless car by German researcher Ernst Dickmanns. Although it was very limited in its capabilities – it could only drive on roads that had no other cars and no passengers – it demonstrated how AI could be useful when applied to real world problems.

IBM's Deep Blue

In 1996, another advancement arrived that broke what was thought to be the theoretical upper limit of machine intelligence. IBM unveiled Deep Blue, a machine that was trained to play chess. Deep Blue played and won against the world chess champion at the time, Gary Kasparov, demonstrating that machine intelligence could match or even outpace human intelligence.

The secret to Deep Blue's success was brute force; trained to calculate millions of possible chess moves in a single second, it was able to win games just by sampling from a vast catalog of moves it could take, estimating the statistical likelihood that each move could lead to winning the game, and choosing the highest-likelihood move from one turn to the next. Deep Blue was a prime example of its time of reactive machine AI, one of the functional categories of AI technology; it was an AI that seemed extremely intelligent on the outside but was just doing what it was designed to do, which was perform high-speed computations on its massive data set of chess moves.

IBM's Watson

In the decades that followed, continued research in AI expanded beyond brute force to explore new strategies for applied machine intelligence. For example, in 2011, IBM released Watson, a machine that used models of natural human language to respond to questions. Watson was portrayed as somewhat of a novelty when it was deployed to play on Jeopardy, winning against long-running champion Ken Jennings. In subsequent years, advancing research in one specific type of machine learning, neural networks, found renewed interest in the AI research community and began to dominate work in the field.



Watch the video below to learn more about Watson.

[Watson and the Jeopardy! Challenge](#)

AlexNet and Alpha Go

This renewed interest is evidenced, for example, by the massive success of the AlexNet project in 2012, the winning submission to the ImageNet Large Scale Visual Recognition Challenge. The success of AlexNet demonstrated a significant leap forward in computer vision, stimulating renewed research in visual recognition systems (owing homage to the early days of the perceptron). Continuing the theme, in 2016, Google's DeepMind lab released AlphaGo, a neural network that used reinforcement learning to train against itself on playing the game of Go.



Watch the video below to learn more about Alpha Go.

Google DeepMind: Ground-breaking AlphaGo masters the game of Go

These examples were major advances in getting machines to do highly complex and ostensibly “intelligent” things. But they were also regarded somewhat as novelties, particularly in the eyes of the public; their purpose was less to do something useful with AI and more to demonstrate that AI was a viable field of research.

The Generative AI Boom

Then, a few years later, something dramatic happened. A new thrust in AI research came into public view with the ascendance of generative AI – artificial intelligence that could not only analyze existing data according to the rules engineered into it, but generate new content based on processes it had learned by itself. These new forms of AI were spurred by advances in deep learning algorithms, which moved AI technology out of the restrictive class of reactive machine AI and into limited memory AI. Suddenly, it seemed as though AI could in fact learn on its own, with minimal human intervention.

In 2020, OpenAI released GPT-3, the third iteration of its generative pre-trained transformers (GPT) technology. GPT-3 represented a major advancement in generative approaches to artificial intelligence, in which machines could be developed to generate new text content based on analyzing massive training data sets of natural language. With the subsequent release of DALL-E in 2021, which could transform text inputs into novel image outputs, and ChatGPT in 2022, which gave the general public access to an AI-driven chatbot that could understand and answer questions with natural language, a new era in AI research had been sparked.

1.7 Traditional and Generative AI

Distinguishing Traditional from Generative AI

Prior to the release of ChatGPT in November 2022, the terms "generative" or "gen" AI were not part of our everyday language. The resulting [unprecedented rise of generative AI](#) changed that in a matter of months. Today, there are [headlines](#) about new advances in generative AI seemingly every week. Despite its now dominant place in our culture, there are still [vastly more applications of the more traditional forms of discriminative \(aka predictive\) AI](#) compared to generative AI. As with discriminative AI, generative AI is a form of machine learning where models learn from data without gaining a conceptual understanding of the underlying ideas. However, in contrast to discriminative models, generative AI is capable of creating new content, such as new text or images, rather than predicting features associated with existing data. Large language models, such as ChatGPT, are types of generative AI that are trained on vast corpora of text and are capable of interacting with humans (and other AI tools) via natural language, i.e., you can chat with them as you would with another person.

What differentiates large language models, e.g., ChatGPT, from earlier attempts at generative AI—and also discriminative AI—is their ability to handle a vast range of tasks. Large language models (LLMs) are typically still performing a kind of auto-complete, but they reference billions of documents when determining a response. These large language models are also heavily adjusted, also known as being fine-tuned, by receiving feedback on their responses from human users (both experts and nonexperts). This kind of human feedback is critical for both the performance of LLMs and reducing the prevalence of harmful outputs.

Nevertheless, LLMs still suffer from what's referred to as "hallucinations" or, as we prefer to call them, errors. Because LLMs, like all AI systems, do not possess a conceptual understanding of the concepts they are discussing, these errors or hallucinations often feel unfamiliar to human users. The [massive costs and persistent hallucinations associated with LLMs](#) have fueled an active debate among AI researchers about whether current approaches will ever lead to AI with more human-like intelligence, much less AGI. The 2025 Association for the Advancement of Artificial Intelligence (AAAI) [Presidential Panel on the Future of AI Research](#) highlighted this debate and showed that >75% of nearly 500 AI researchers believed that simply scaling existing approaches would not lead to AGI.

How Does Generative AI Fit Into the AI Landscape?

Why can ChatGPT generate essays, poems, or code from a single prompt? Why are some responses accurate while others contain difficult-to-catch errors that we call hallucinations?

In this lesson, we will unpack what generative AI is, how it differs from more traditional forms of AI, and how what makes these systems capable also makes them unpredictable and prone to errors. We'll explore the architecture behind tools like large language models, the role of transformers and attention mechanisms, and how generative AI models fit within the larger AI landscape.

Machine Learning and Generative AI

One of the things that's important to understand about machine learning is that it is a subfield of artificial intelligence. With artificial intelligence, we say that it's a program with the ability to simulate human intelligence. That's obviously a very broad concept and something that we're not really going to get into today.

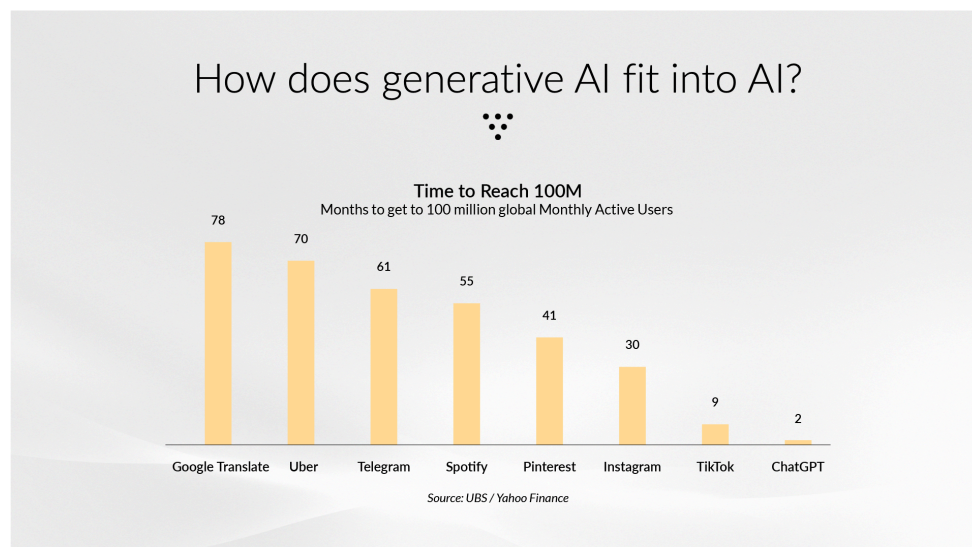
However, if we think a little bit more narrowly, one aspect of intelligence is the ability to learn information from data, and that's what we call machine learning. Machine learning refers to programs with the ability to learn functions—these are mathematical objects that map things onto each other—or parameters of those functions from data. Don't worry, we'll make this less abstract in the coming slides.

As we get further into the discussion of machine learning, we'll talk about how generative AI is actually a subset of machine learning.

The Rapid Adoption of Generative AI

One of the things that's important to think about in the context of generative AI is how rapidly it's been adopted.

If you look at the chart here, we can see some recent technology innovations, things like Uber, Spotify, and Pinterest, and we're looking at the time it takes for these apps to reach 100 million users. You can see for Uber, it was 70 months—multiple years—before Uber had 100 million users.



For ChatGPT, it was only two months before it reached 100 million users—absolutely unprecedented. If you felt like it's been a bit of a whirlwind over the past few years, it really has been unprecedented in terms of the speed of adoption.

What Is a Generative Model?

One example of what most previous machine learning or AI models were trying to do, is what we would call **discriminative** or sometimes **predictive AI**. In this case, what we want to be able to do is recognize a picture, recognize a painting of a woman.

Whereas for generative AI, we actually want to create a picture. We want some kind of tool that can generate a new image. More precisely, we need to model the joint distribution over the data that was used to train the model that itself has been assigned with labels.

Imagine we have lots of pictures and it says, woman, dog, cat, fried chicken, and we're going to use those data to train our generative model.

Neural Networks: Encoders and Decoders

Most of these generative AI models use neural networks. But they use a new class of neural networks that previously was somewhere between uncommon and invented, depending on exactly where we want to draw the line between when these things were developed.

A way to think about these neural networks: they have different components called **encoders** and **decoders**. The encoder does what it sounds like it would do—it takes text, like "nice to meet you," and encodes that in a mathematical or data-driven representation, the kind of thing that a computer will be able to understand. Then the decoder takes that vectorized or mathematical representation of the text or the image and translates that back into a text-based output or an image-based output.

We've shown an example here of going through a translation task where we're going to encode the text in one language, run it through a translation algorithm, and then decode it back into a second language.

The Transformer and Attention Mechanism

The key invention for most of what we use with respect to generative AI is something called the **transformer**. Inside that transformer was a new kind of way of organizing data that's referred to as the **attention mechanism**. This is what led the authors of the paper to write in the title, "Attention is All You Need," arguing that this kind of attention mechanism was critical for advancing the capability of generative AI.

The analogy for thinking about how the attention mechanism works is to imagine yourself in a crowded room, in a party, in a restaurant where it's very loud. You can hear people talking at the table next to you. You can hear the server taking an order. You can hear someone in the kitchen dropping a plate. You need to focus on the person who is across the table from you that's telling you a story.

That's how the attention mechanism works. What it tries to do is learn the contextual relationship and identify the types of information that's important to pay attention to in the conversation that you're having. If you want to respond to the thing that the person is telling you from across the table, you probably don't want to be paying very much attention to what's going on at the table next to you or going on with that plate that's crashing in the kitchen.

Semi-Supervised Learning

Another thing that's a little bit different about generative AI is that it's trained in what we call a **semi-supervised** fashion. Again, inside of this field of artificial intelligence, we talked about how machine learning—programs that can learn functions or parameters from data—is a subset.

Generative AI is a subset of machine learning, so we're still training models to learn parameters, to learn functions from data. Normally within machine learning, we would think of unsupervised versus supervised learning:

- **Unsupervised learning:** Our data don't have labels. We just have a whole bunch of pictures but no one has told us what's in the pictures.
- **Supervised learning:** We have a whole bunch of data where people have labeled what's inside of the pictures—cat, dog, woman, fried chicken, waiter dropping a plate back in the kitchen.

Large language models are typically trained by a form of semi-supervised learning, which means that the model itself is actually engaging in part of the labeling process. This is one thing that's very important to remember: a lot of what goes on in AI relies on human-labeled data.

But in some cases, we have a machine that is partially engaged in data labeling or evaluating the machine's output. This raises questions about the potential quality of the training model if we have no humans in the loop that are a part of the training process.

One of the things that we want to pay attention to when we're evaluating a generative AI model is how it was trained, how much human supervision was involved, how much human-labeled data was involved, versus how much self-supervision was occurring on the part of the machine learning algorithm or what we would refer to as the large language model.

Hallucinations and Errors

One of the things that you've probably heard about is that these large language models or generative AI tools hallucinate.

One of the problems with AI tools is that when they make mistakes, they tend to make them in unhuman-like ways, which can be very uncomfortable for us. The word "hallucination" sounds like a thing that a human would do, and these errors are very nonhuman-like.

A kind of hallucination in the context of, say, cybersecurity would be: you have an AI assistant, a large language model, that is detecting threats that aren't there. They're making mistakes with respect to their threat assessment, which, of course, could be a significant problem.

The way that we refer to this is that it produces errors instead of hallucinating. This is what has led many people to refer to large language models as **stochastic parrots**, meaning that they don't really understand what they're doing. They're just performing autocomplete. They can answer a question without really understanding the meaning. They can just do pattern recognition and provide an output.

Video and Image Generation

A slight difference is with video and image generation. Video and image generation, for the most part, work by adding noise to images and then training models to denoise those images. They're slightly different from this kind of autocomplete, stochastic parrot of most of the text-based or generative AI tools. It is important to remember that as this field evolves very rapidly and we see new kinds of generative AI models, they are shifting the paradigm a bit.

Not every generative AI model is going to be the same kind of stochastic parrot. They may have different kinds of ways that they're being trained. But importantly, these models still don't have an understanding of what's going on. They're still doing pattern recognition and memorization.

Agentic AI

Lastly, **agentic AI** has come on the scene, again, similarly quickly with respect to the original ChatGPT. In a matter of months, we went from nobody talking about agentic AI to everybody talking about agentic AI.

It's helpful to think about where this fits in the context of generative AI:

- **Generative AI models** (like ChatGPT): Create new kinds of content by learning from data. We can use them to write code and summarize documents.
- **Agentic AI**: Can plan and execute tasks where they've been trained, usually by a process of reinforcement learning, to work in a way that's quite independent from humans. This is something that we want to keep in mind as we're looking at these agentic AI solutions—how much human time was put into training them.
- **Multi-agent systems**: Multiple agents may be strung together with different kinds of large language models performing fairly complicated tasks. This is where many people think the future is going. But it's very early days for this. We want to make sure that we're going to wait and see a bit, evaluate where these solutions are before we jump in with some of these agentic AI tools, as they're still fairly new.

Traditional AI Examples

As mentioned previously, more traditional kinds of AI, i.e., discriminative or predictive ML models, are still the most used forms of AI. While these traditional models cannot generate new content and, most importantly, lack the more generalizable ability to interact with humans via chat, they can make useful predictions and identify key differentiators between data. Whether you realize it or not, traditional AI is a major part of our daily lives. As an example of traditional AI, consider recommender systems, which drive various platforms, including social media, music streaming services, and movie and TV subscription services. For example, a music subscription platform like Spotify uses your listening history to determine your musical preferences and then recommends new music based on that information.

Similarly, movie and TV subscription services, such as Netflix and Hulu, use your viewing history to model your viewing preferences and then encourage you to consume more media through these platforms by offering content you are likely to enjoy. In these cases, recommender systems are traditional in that they are reactive to data about your behavior and try to predict whether you will respond positively to recommendations. Usually, these models are designed to do one thing, e.g., to analyze your listening

patterns and predict preferences based on those patterns, or they might analyze the films you've enjoyed and suggest others that will keep you on the platform. They are not generative, i.e., they aren't creating new music or new movies based on your preferences, and they can't do anything else like suggest a recipe for a meal to cook that would go well with what you're listening to. They recommend—and that's it. (That said, we are already seeing efforts where generative AI tools are providing such services, [such as generating entirely new media based on what you want to see!](#))

Generative AI Examples

Traditional AI examples are plentiful, and they are present in places you may not expect or notice, e.g., movie and song recommendations. In contrast, generative AI examples are much smaller yet far more present in the media and culture. Examples include tools such as OpenAI's [ChatGPT](#) (for text generation) or [DALL-E](#) (for image generation), [Google Gemini](#), graphics editing and generation features in software such as [Adobe Photoshop](#), creative tools such as [Midjourney](#), and other more general-purpose tools such as Microsoft's [Copilot](#). These newer tools are called "generative" because they are capable of generating new content rather than iterating upon existing content or data.

One common subcategory of generative AI is chatbots, which work through large language models (LLMs). These models are trained on enormous amounts of text data and use that data to create new content, such as responses to questions asked through a chatbot. Another subcategory includes image generation tools such as DALL-E; such tools operate similarly, but instead of generating new text content, they take text inputs and generate new graphic or image content from those inputs. In the following section, we provide more details on types of generative AI models.

1.8 Overview of Generative AI Modalities

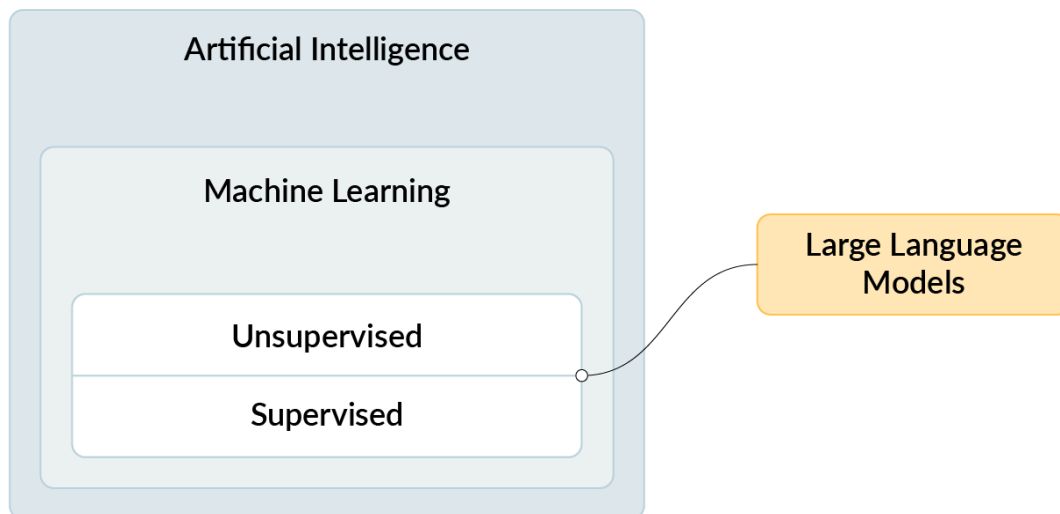
Generative Models

In this section, we will delve into more detail about generative AI models, including large language models (LLMs). We will see how LLM “hallucinations” (which we prefer to call errors) have inspired the development of many advances in generative AI, including retrieval augmented generation (RAG) and agentic AI. In future sections, you will learn how agentic AI has expanded to cover many additional use cases and push the boundaries of what’s possible with both traditional (i.e., discriminative) and generative AI.

Generative AI is now used across a wide range of applications, including the following:

- Text generation produces realistic and coherent text for purposes such as writing stories and articles and engaging in chatbot conversations (e.g., ChatGPT, Claude, Gemini, BERT, etc.). Most text generation AI models utilize transformers, but not all text generation methods are based on LLMS.
- Image generation involves creating images from text prompts or manipulating existing images (e.g., DALL-E and Stable Diffusion). Most image generation AI models use diffusion models.
- Code generation writes code from plain text descriptions (e.g., Copilot, CodeWhisperer, and models such as ChatGPT, Claude, and Gemini). Most code generation AI models use transformers and are LLMS.
- Video generation creates videos from text prompts or other inputs (e.g., Sora and Veo). Most image-generation AI uses diffusion models.

Modern LLMS such as current versions of ChatGPT, Claude, and Gemini, are trained on huge datasets (typically text) and are used for a range of tasks, including generating text and code.



<Alternative text: Machine learning is a type of AI, and LLMS are a type of machine learning.>

LLMs are built using deep learning models (i.e., multi-layer neural networks) and rely primarily on the transformer architecture for their enhanced performance compared to earlier generative ML models used for text generation. Earlier in the module, we discussed the transformer—the type used today was introduced in 2017—which uses what is called the attention mechanism to focus on the important relationships between entries in a sequence of data, such as a sentence or a news article.

What we did not discuss is that transformers also enable the parallel processing of sequential data (such as words in a sentence or heart rate values over time), allowing models to learn long-range dependencies and complex relationships in massive datasets efficiently. This ability to parallelize computation leverages graphics processing units (GPUs) and explains why LLMs can be trained using such massive datasets. The combination of the attention mechanism and the ability to parallel process data has led to significant improvements in the quality of generated text and code by LLMs.

However, as you're likely aware, LLMs produce lots of errors, which are typically referred to as hallucinations. Hallucinations refer to LLM outputs that are factually incorrect, nonsensical, or unrelated to the prompt. You've all encountered hallucinations or read about some of the [really egregious ones](#) online. While the ability of LLMs to generate human-like text has increased dramatically, their tendency to hallucinate hasn't gone away and raises concerns about their reliability and trustworthiness. These hallucinations, i.e., errors, should be top of mind for anyone considering using LLMs.

 False Citations AI creates academic papers, books, or studies that don't exist but sound credible and authoritative. <i>AI Output: "According to Johnson et al. (2023) in the Journal of Advanced Studies..."</i>	 Invented Facts Generating statistics, dates, or historical events that never happened but seem plausible. <i>AI Output: The Eiffel Tower was built in 1887 and is 324 meters tall (close, but wrong year)</i>
 Fictional People Creating detailed biographies, quotes, or achievements for people who don't exist. <i>AI Output: Dr. Sarah Mitchell, renowned physicist at Stanford, discovered...</i>	 Impossible Combinations Mixing real concepts in ways that don't make logical or physical sense. <i>AI Output: The quantum computer uses photosynthesis to process data faster</i>

<Alternative text: Overview of different types of hallucinations from AI models: false citations, invented facts, fictional people, and impossible combinations of information>

The underlying cause of hallucinations is rooted in the way LLMs model the "world." These models learn statistical patterns in vast amounts of text data without developing an understanding of its meaning. When prompted to generate text, LLMs essentially predict the most probable sequence data, e.g., words, based on their training data, without verifying the factual accuracy or logical coherence of the output.

This stochastic (i.e., property of being well-described by randomness) process can lead to the generation of plausible-sounding but incorrect or nonsensical statements.

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Watch the following video to explore why LLMs hallucinate:

[Why Large Language Models Hallucinate](#)

Stochastic Parrots

The concept of a “stochastic parrot” is used to describe this LLM feature of outputting information you don’t understand based on a prompt. LLMs, like parrots, can mimic human language without comprehending the underlying meaning of the words they are saying. While LLMs can string together words and phrases that statistically fit the context, the models lack the ability to reason, fact-check, or critically evaluate the information being generated.



In the video below you will learn more about the dangers of stochastic parrots from a panel of experts. You are not required to watch the full video. Use the chapter feature in the description of the video to further explore at least one topic that interests you. The chapter feature is housed within the video description.

[On the dangers of stochastic parrots: Can language models be too big?](#)

Reducing Hallucinations and Improving LLM Output

The only certain way to evaluate AI output for quality and errors (i.e., hallucinations) is for an expert to manually inspect the generated text. However, checking LLM output for factual accuracy and logical coherence quickly becomes a time-consuming process, especially for large volumes of text, and it requires human evaluators with a high degree of expertise in the topic. Increasingly, researchers are developing automated methods to improve LLM output, which reduce errors or enhance the detection of them. Because these methods often involve analyzing the statistical properties of the text to identify potential AI-generated hallucinations, they will always be imperfect.

Broadly, we can group the proposed methods for reducing hallucinations and improving the quality and rigor of LLM outputs into the following six categories:

1. **Prompt engineering:** The most common and effective approach for improving the output of LLMs is to refine the input prompt used to generate the response. The best prompts are those that provide very specific instructions, even including the desired output format of the response. For most LLM use cases, it is considered best practice to try prompting the model in different ways and then compare the outputs for similarity.



You can learn more about prompt engineering in this [CSA course](#) on the topic.

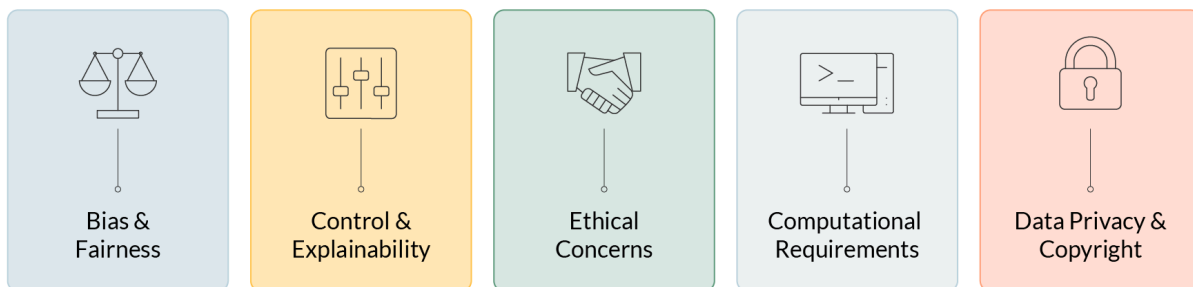
2. **Retrieval augmented generation (RAG):** Linking an LLM to a database—or corpus of documents, such as papers curated from peer-reviewed scientific literature—can help reduce hallucinations by specifically requiring the LLM to return chunks of text contained in the linked database or corpus. Adoption of RAG models is becoming widespread and presents an exciting path forward for LLMs. However, RAG models can still hallucinate (i.e., produce errors).
3. **Improving training data:** By constructing and curating training datasets that include more reliable sources, researchers believe we can decrease the chance that LLMs will learn misinformation. These LLMs trained on “better” or more rigorously vetted training data are hypothesized to produce fewer hallucinations. However, this hypothesis is not proven, and this approach doesn’t address the underlying issue associated with how LLMs model the world or their lack of understanding.
4. **Fact verification techniques:** Incorporating fact-checking into LLMs is an active area of research that shows promise for reducing hallucinations (at least when human experts are used as fact checkers). This fact-checking or verification involves cross-referencing LLM-generated output with external knowledge bases or with experts to evaluate its accuracy. Companies typically use vast numbers of human fact-checkers to train and evaluate LLMs. Increasingly, LLMs are being used as fact-checkers for other LLMs, which presents opportunities and raises new challenges associated with trust and explainability.
5. **Reinforcement learning with human feedback (RLHF):** Training LLMs with human feedback can guide models toward generating more accurate responses. However, current RLHF methods require evaluating hundreds or even thousands of responses, which can quickly become

prohibitively expensive and raise ethical concerns around the compensation provided to human evaluators.

6. **Human Experts:** Because no method is foolproof, it is impossible to determine whether an LLM's output is correct by any means other than direct assessment by a qualified expert. As a result, we believe the future of LLM use in high-risk areas, such as in the health sciences, will always require the involvement of human experts.

Associated Challenges

As with all AI models, today's generative AI (including advanced RAG and agentic systems) also suffers from challenges associated with generalizability (largely derived from overfitting) and explainability. However, using generative AI models raises challenges not unique to generative AI that are made worse by discriminative ML models.



- **Bias, Fairness, and Transparency:** Generative models will reflect biases in the training data, potentially leading to discriminatory outputs. These bias issues impact both discriminative and generative AI, but evaluating generative ML tools for biases and fairness often requires novel approaches compared to discriminative ML tools. There are two main reasons it can be harder for us to evaluate generative models for bias and fairness: First, we are less familiar with them, and second, they are typically less “explainable” than discriminative models. As a result, transparency in the underlying data, training procedures, and model architectures is important to consider when evaluating generative models.
- **Control and Explainability:** Controlling the output of generative models and understanding their decision-making processes is still largely a black box. Again, such explainability issues are not unique to generative ML, but they are made worse by the high degree of model complexity. What is somewhat new with generative ML models is control, or being able to predict what output will be returned for a given input. While we observed that neural networks performing discriminative modeling also face this kind of control issue, we simply didn't label it as such.
- **Ethical Concerns and Discrimination:** The potential for misuse of generated content, such as deepfakes and the spread of misinformation, raises new ethical concerns for generative ML models. Additionally, whether intended or unintended, concepts like social scoring can lead to illegal forms of discrimination.
- **Computational Requirements:** Training and running most modern generative models requires substantial computational resources. Although some debate the impact of the carbon, water, and

electrical demands of generative AI, there is no doubt these models require vast computational resources to train and operate.

- **Data Privacy and Copyright:** Again, even with discriminative ML models, we must comply with data ownership, privacy, and copyright regulations and norms, issues that are exacerbated by the rise of generative AI. However, given the massive data scales associated with generative AI, and the uncertain legal and regulatory landscape, there are unique and fast moving challenges associated with generative AI, data privacy, and copyright considerations.

1.9 Cloud Security Models

IaaS, PaaS, and SaaS

Nearly every user of generative AI interacts with these models via the cloud. In this section, we briefly define the most common cloud service models and discuss trade-offs at each level (control vs. complexity vs. accessibility). We provide examples of services at each level to help you better understand how generative AI is typically deployed and accessed in the real world. Cloud service models are typically subdivided into three tiers (IaaS, PaaS, and SaaS) based on the level of hardware or software use and degree of customization afforded to clients and end-users. Because organizations may have conflicting needs across teams, a shared service model is often adopted that centralizes key aspects associated with security, while flexibility is still provided to individual business units. Regardless of the model selected, individuals, teams, and organizations all have a shared responsibility and accountability associated with data and cyber security.

Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is a cloud computing model where clients lease or rent access to physical and virtual infrastructure such as GPUs, servers, and networking. Users then access these computing resources virtually. Clients using IaaS do not need to maintain rented infrastructure; however, they usually have to manage most (or all) of the necessary software, including operating systems and applications. Organizations must secure the entire software stack, protect training datasets and model weights from theft; implement network isolation for training infrastructure; and manage access controls for compute resources. Most major cloud computing vendors offer IaaS solutions, such as Amazon EC2, Google Compute Engine, and Microsoft Azure Virtual Machines, with numerous options also available from smaller providers.

Platform as a Service (PaaS)

Building from IaaS, Platform as a Service (PaaS) includes rented or leased software tools along with physical and networking infrastructure. While clients must manage most higher-level applications and data, PaaS providers typically manage runtime, middleware, operating systems, virtualization, servers, storage, and networking. While providers will handle most aspects associated with infrastructure security, users must manage things like API keys, implement security for model endpoints, monitor for anomalous usages, and ensure compliance with data storage requirements. As a result, PaaS allows developers to create customized applications without needing to manage as much of the physical infrastructure and lower-level software tasks as IaaS models. Examples include Google BigQuery, Heroku, and Azure App Service.

Software as a Service (SaaS)

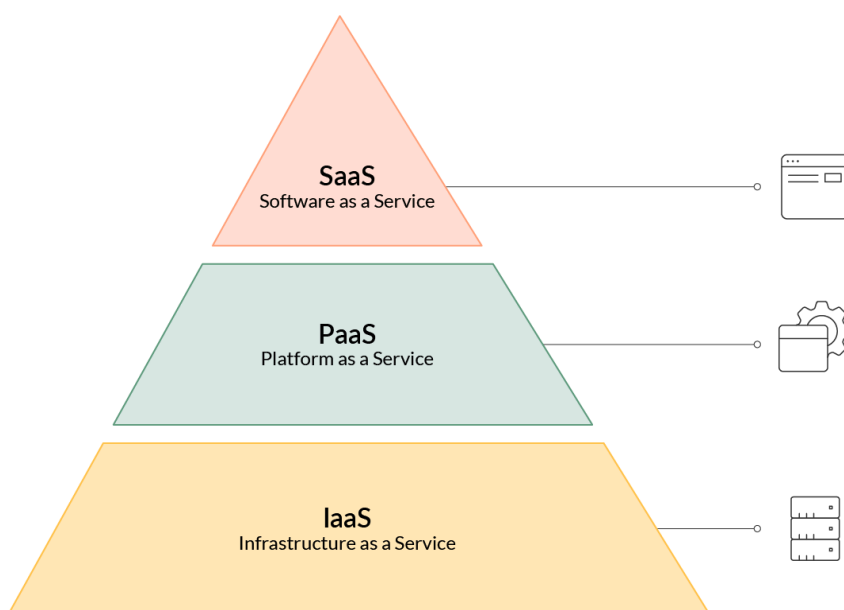
Finally, Software as a Service (SaaS) solutions deliver fully functional applications to end users, usually including access to all necessary infrastructure and networking. This model typically requires less management by clients compared to IaaS and PaaS. Under SaaS models, the provider normally manages the entire technology stack, including infrastructure, lower-level software, and high-level applications. Users access the software through web browsers or APIs. Many of the security aspects are handled by the SaaS providers, but users must still protect sensitive data in prompts, implement usage policies to prevent the generation of harmful content, ensure compliance with data privacy regulations, and manage the risk

of AI-generated misinformation or intellectual property violations. Additionally, the specific terms, level of maturity, and other trust-based details associated with the SaaS provider must be considered when selecting a solution or deciding to use this higher-level tier than IaaS or PaaS. Also, SaaS options are almost always less customizable than IaaS or PaaS. Examples include Google Workspace, Salesforce, and OpenAI's ChatGPT interface.

In summary, the three levels of cloud service provide distinct pathways for implementing and using generative AI, ranging from IaaS, which offers raw computational resources for training massive models, to PaaS, which provides managed development environments with pretrained model access, and to SaaS, which delivers ready-to-use AI applications directly to end users.

Each level presents unique cyber security considerations: IaaS requires organizations to secure their entire stack from operating systems upward while protecting sensitive training data and model weights, PaaS shifts some security responsibility to providers but requires careful API key management and access control for fine-tuned models, and SaaS minimizes user security burden but requires trust in the provider and raises concerns about data privacy, prompt injection attacks, and the potential exposure of sensitive information through AI interactions.

Organizations must balance their need for customization and control with the need for security expertise and resources when selecting the appropriate generative AI service model. As a result, many organizations adopt a shared services model where key aspects are centralized. While shared services models can create bottlenecks that reduce operational efficiency, these bottlenecks can be offset with security and efficiency gains. Regardless of the model chosen, individuals, teams, and organizations share a responsibility to ensure that data is secured and used appropriately.



<Alternative text: Levels of cloud service models: At the base is infrastructure as a service. Platform as a service is in the middle. Software as a service is at the top.>

1.10 Summary

Key Concepts Mastered

In this module, you developed a stronger foundation in AI, especially modern approaches to generative AI. You're now ready to move on to modules that dive deeper into the security and cloud deployment challenges associated with AI systems.

Understanding AI Fundamentals

At its core, AI represents systems capable of performing tasks associated with naturally intelligent, historically human systems. While natural intelligence remains ill-defined and debated, we believe that intelligent systems can perceive aspects of their environment, respond flexibly, and learn from experience. Many AI systems exhibit these features of intelligence, but most are still relatively narrow in their capabilities. Even LLMs perform by memorizing, and they lack the conceptual understanding of ideas associated with human intelligence. As a result, artificial general intelligence (AGI) remains a future goal, and whether or not we can ever obtain AGI is actively debated.

The Generative AI Revolution

We are witnessing a dramatic shift in AI, moving from discriminative AI, which analyzes and classifies existing data, to generative AI, which creates entirely new content. This revolution leverages five core architectural types that work in different ways to generate content. Autoregressive models, the oldest form, generate data sequentially based on prior information. Generative adversarial networks (GANs) use competitive generator-discriminator networks to create realistic content. Variational autoencoders (VAEs) employ collaborative encoder-decoder systems to learn and generate from data representations. Transformers, which utilize attention-based mechanisms, power modern large language models, such as GPT and BERT. Finally, diffusion models, which are trained to remove noise that has been gradually added to data, are the most advanced image and video generation models.

Historical Context Matters

While the term "AI" was coined at the famous 1956 Dartmouth Conference, the foundations of AI extend back into the earliest days of modern science. Key milestones following the 1956 conference include the development of perceptrons in 1958, the introduction of transformers in 2017, and the public release of ChatGPT in 2022. Modern AI success resulted from a perfect combination of converging factors: increased computational power through GPUs originally developed for gaming, abundant data from the internet and connected devices, and crucial algorithmic breakthroughs that made it possible to process this information at scale.

LLMs: Power and Limitations

Large language models have revolutionized natural language processing through the transformer architecture and its associated attention mechanism. The transformer facilitated the kind of parallel processing necessary to achieve massive scale. However, despite their impressive ability to respond thoughtfully to a wide range of questions, search the web, and generate code, these models lack a conceptual understanding of the topics they discuss. This lack of conceptual understanding is what led

researchers to describe LLMs as "stochastic parrots," as they generate outputs based on statistical patterns rather than true understanding. As a result, LLMs generate plausible-sounding but sometimes incorrect information, known as hallucinations. To mitigate hallucinations, researchers have developed various approaches, including prompt engineering, retrieval augmented generation (RAG), expert fact verification systems, reinforcement learning with human feedback (RLHF), and agentic AI. Nevertheless, hallucinations are inevitable when using LLMs.

Cloud Deployment Models

Many of today's AI tools are trained and deployed using cloud services. As a result, understanding the various cloud service models is now a critical aspect of anyone's AI education. In this module, we discussed the three main cloud service models—IaaS, PaaS, and SaaS—and shared service approaches. Determining the ideal combination of service models, including a shared services approach, requires balancing trade-offs between control, customizability, and management complexity. Regardless of the service model, the proper use of data in the cloud is the responsibility of everyone.

Looking Ahead: Your Continued AI Journey

Your foundation in AI concepts is just the beginning. The field is rapidly advancing, and staying current requires ongoing learning and hands-on practice.

Recommended Next Steps

Continue through the course modules. In future sections, you will learn much more about the use of AI in modern cloud environments. You should also target opportunities to test your knowledge with hands-on implementation. Such hands-on training might look like starting a new PoC inside of your organization or developing toy models at home. Remember that each new AI system introduces new types of vulnerabilities—from training data poisoning to model inversion attacks—that require updated security controls and monitoring strategies tailored to your organization's risk profile and regulatory requirements.

Security-First Implementation

As you deploy AI systems, you should strive to maintain a security-first approach. While security-first approaches are not new, AI is evolving into a critical piece of infrastructure. As a result, AI security encompasses both traditional cyber security concerns and novel threats often particular to AI. Your role as security professionals is crucial in ensuring that AI systems enable innovation within acceptable risk parameters.

Change Log

Version #	Approved by	Notes
20251015	HMR	Initial release